

# LuSIR: Latent Upscaling via Self-trained Image Restoration without T2I Pretraining

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Snapshot: formal x4 benchmark complete, Stage 2 clean-fidelity learning-rate probes complete, and a separate high-frequency residual diffusion path selected as the next generative experiment.

`paper/TECHNICAL_REPORT.md` is the canonical report source. `paper/sr_diffusion_report.pdf` and `paper/main.tex` are generated from it with `paper/build_report.sh`, so the Markdown and PDF contain

the same report.

## 1 Objective

LuSIR trains a vision-only x4 latent diffusion super-resolution model without using a pretrained text-to-image backbone. The active task is:

LR 128x128 -> HR 512x512

The model path is:

```
HR image -> factor-4 VAE -> HR latent
LR image -> condition encoder -> condition latent
noisy HR latent + condition latent + timestep + domain id -> conditional U-Net
denoised latent -> VAE decoder -> SR output
```

The numbered stages describe training order, not a mandatory Stage 1 -> 2 -> 3 -> 4 runtime chain. The current runtime choices are:

public Colab default:

LR -> Stage 2 XL condition encoder -> residual refiner v2 -> Stage 1 decoder

current detail research candidate:

LR -> dual-context LSDIR Stage 2 -> Stage 1 decoder -> detail branch v1d

generative comparison:

LR -> Stage 2 condition encoder -> Stage 3 OR Stage 4 diffusion U-Net  
-> Stage 1 decoder

Stage 4 is a Stage 3-derived replacement diffusion checkpoint. It does not run after Stage 3. Planned Stage 5 distillation would similarly replace the slower diffusion sampler rather than append another serial module.

## 2 Data

The base photo100k split has 103,450 training images and 100 fixed validation images. The completed dual-context Stage 2 scale-up adds 30,000 unique LSDIR training images, for 133,450 unique training images and the unchanged 100-image validation set. LR inputs are generated on the fly from HR crops. Earlier XL work used `photo_v3_noise_mix`, a strong denoise-focused curriculum with no clean share and 80% combined v2/v3 cases. Later work introduced `photo_detail_mix`: 35% clean, 48% detail-preserving photo degradation, 15% mild degradation, and 2% strong `photo_v2`. This keeps a small robustness tail while shifting the primary objective toward user-facing detail restoration.

## 3 Completed Baselines

| Stage                | Run                                         | Result                                       |
|----------------------|---------------------------------------------|----------------------------------------------|
| Stage 1 VAE          | autoencoder_photo10k_b16_eval_online        | eval/psnr 40.19                              |
| Stage 2 photo100k    | latent_pretrain_photo100k_b64               | best latent loss 0.21230                     |
| Stage 3 photo100k    | diffusion_photo100k_b32                     | sampled val100 25.3745 PSNR                  |
| Stage 4 photo100k    | diffusion_photo100k_b32_stage4_condition    | sampled val100 25.4072 PSNR                  |
| Stage 4 photo100k v2 | diffusion_photo100k_b32_stage4_condition_v2 | sampled val100 22.8426 PSNR, +0.4323 over bi |

The v2 task has a stronger degradation distribution, so its absolute PSNR is not directly comparable with the earlier mild photo100k run.

## 4 Exploratory Strict-Bicubic Five-Image Comparison

The ordinary LuSIR validation metrics use task-specific degradations, so their absolute PSNR values should not be compared directly with published bicubic-only SR benchmarks. A small strict-bicubic diagnostic was added to separate degradation difficulty from reconstruction capacity. The new `benchmark_bicubic` preset applies only PIL bicubic factor-4 downsampling and does not add blur, noise, compression, color changes, or sharpening.

The shared diagnostic protocol is:

```
source images: DIV2K validation 0801-0805
HR input: deterministic center 512x512 crop
LR input: strict PIL bicubic x4 downsample to 128x128
metric: mean per-image RGB PSNR over the full 512x512 crop
deterministic inference: CPU FP32
diffusion inference: condition initialization, start timestep 50, 32 DDIM steps
```

Loaded parameter counts include the entire 21.10M-parameter Stage 1 VAE because the current runners load the full module, although inference executes only its 10.52M-parameter decoder. This matches the existing 509.658M Stage 4 XL accounting.

| Inference path                   | Selected checkpoint | Loaded params | Mean RGB PSNR | vs bicubic |
|----------------------------------|---------------------|---------------|---------------|------------|
| Bicubic                          | n/a                 | n/a           | 29.5999       | n/a        |
| Stage 2 XL condition-only        | step 72000          | 40.040M       | 30.5677       | +0.9678    |
| Stage 2 XL + residual refiner v2 | step 39000          | 48.106M       | 30.8205       | +1.2206    |
| Stage 2 multiscale               | step 46000          | 76.591M       | 31.6068       | +2.0069    |
| Stage 2 dual-context LSDIR       | step 98000          | 140.334M      | 31.7411       | +2.1412    |
| Dual-context + detail v1b        | step 39500          | 141.675M      | 31.8135       | +2.2136    |
| Dual-context + detail v1c        | step 6000           | 141.689M      | 31.8154       | +2.2155    |
| Dual-context + detail v1d        | step 99500          | 143.354M      | 31.9513       | +2.3514    |
| Stage 4 XL edge, 32-step sampled | step 4250           | 509.658M      | 29.5487       | -0.0512    |

The strict-bicubic result confirms that LuSIR already operates in the 30-32 dB range when the LR input is not additionally corrupted. The much lower `photo_detail_mix` and strong-preset values primarily reflect degradation difficulty rather than a direct 6-7 dB gap to bicubic-only SR papers.

The capacity trend is also informative. Stage 2 XL to multiscale gains +1.0391 dB, and multiscale to dual-context gains another +0.1343 dB. The detail branches then provide smaller but consistent gains over dual-context: v1b +0.0725 dB, v1c +0.0744 dB, and the completed 3.02M v1d +0.2102 dB. V1d improves v1c by +0.1358 dB after three epochs. This confirms that the long capacity run was useful, but the visual change remains conservative and is not a perceptual-detail breakthrough.

The 509.7M Stage 4 XL edge checkpoint is the clearest counterexample to treating parameter count as a quality ranking. On this clean bicubic diagnostic it falls -0.0512 dB below bicubic and -1.0190 dB below its own Stage 2 XL condition-only path. Visual inspection shows that it can add apparent sharpness, but it also changes clean input details enough to reduce reconstruction fidelity. This is consistent with its strong-noise denoise/color-cleanup training role; it should not be routed unconditionally for clean inputs.

This diagnostic is deliberately small and is not a published-benchmark claim. It uses five center crops, RGB PSNR, no border shave, and PIL bicubic rather than the full-image Y-channel/Matlab-bicubic conventions used by many SR papers. The formal full-image benchmark described next supersedes it for clean-bicubic fidelity comparison.

The machine-readable snapshot is stored in:

```
metrics/benchmark_bicubic5_lusir_model_comparison.json
```

## 5 Formal Full-Image x4 Benchmark

A formal evaluator now measures full-image DIV2K validation, Set5, Set14, and Urban100 using their public x4 bicubic LR pairs. It uses MATLAB-compatible BT.601 Y conversion, a four-pixel border shave, and MATLAB-style

SSIM. All candidate outputs are required to match the HR dimensions exactly; the evaluator never hides an error by resizing an output.

| Candidate               | DIV2K          | Set5           | Set14          | Urban100       |
|-------------------------|----------------|----------------|----------------|----------------|
| Bicubic                 | 28.1044        | 28.4318        | 26.0928        | 23.1412        |
| RealESRNet x4plus       | 28.8250        | 29.3828        | 27.1465        | 24.4613        |
| RealESRGAN x4plus       | 26.6125        | 26.6160        | 25.4216        | 22.6709        |
| SwinIR classical x4     | <b>31.0838</b> | not run        | not run        | not run        |
| LuSIR refiner v2        | 28.7857        | 28.1896        | 27.3704        | 24.9176        |
| LuSIR dual-context base | 29.9575        | 31.6621        | 28.2441        | 25.4816        |
| <b>LuSIR detail v1d</b> | <b>30.1602</b> | <b>31.8892</b> | <b>28.4123</b> | <b>25.8755</b> |

The table reports Y PSNR; full Y SSIM and RGB PSNR values are preserved in the machine-readable results. V1d improves the frozen dual-context base on all four datasets. Its Y PSNR gains are +0.2027, +0.2271, +0.1682, and +0.3939 dB, respectively. Its Y SSIM is 0.83421, 0.89440, 0.77998, and 0.77875. This validates the branch redesign under a standard full-image protocol, with the strongest gain on texture-heavy Urban100.

The official SwinIR classical x4 checkpoint reaches 31.0838 / 0.85228 on the same DIV2K evaluator. It is +0.9235 dB Y PSNR and +0.01807 Y SSIM ahead of detail v1d. The detail branch is doing useful corrective work, but this remaining gap identifies the Stage 2/base reconstruction path as the primary clean-fidelity bottleneck.

RealESRNet and RealESRGAN target real-world degradation and perceptual quality, so their clean-bicubic fidelity scores do not establish a general visual quality ranking. Likewise, beating these checkpoints under this protocol does not establish classical-SR SOTA. A classical fidelity baseline, perceptual metrics, real-degradation evaluation, and blind human review remain necessary. The reproducible protocol and commands are in `docs/SR_BENCHMARK.md`; full machine-readable results are in:

```
metrics/formal_x4_benchmark_lusir_realesr_summary.json
metrics/formal_x4_benchmark_lusir_realesr_metrics.csv
metrics/formal_x4_benchmark_div2k_swinir_summary.json
metrics/formal_x4_benchmark_div2k_swinir_metrics.csv
```

## 6 Stage 2 Clean-Fidelity Continuation and Learning-Rate Probes

The clean-bicubic continuation starts from dual-context Stage 2 best98000 and uses `benchmark_bicubic` with a PSNR-oriented decoded loss balance. Its task-specific val100 decoded-PSNR proxy improved gradually:

| Step  | Decoded PSNR proxy |
|-------|--------------------|
| 1000  | 25.019             |
| 4000  | 25.031             |
| 9000  | 25.045             |
| 15000 | <b>25.057</b>      |
| 17000 | 25.054             |

These values are not directly comparable with the formal full-image Y-channel results. The formal DIV2K comparison remains LuSIR detail v1d at 30.1602 dB and SwinIR classical x4 at 31.0838 dB.

Separate learning-rate probes preserved the original step-15000 checkpoint. A 20x continuation collapsed to 15.72 dB at its first evaluation. A 5x continuation remained below the original run, while a 5x from-initialization run reached 25.033 dB at step 4000 versus 25.031 dB for the original learning rate. The difference is too small to treat as a gain. The original 5e-6 learning rate is retained, and learning-rate scarcity is rejected as the main bottleneck.

The result also clarifies the objective boundary. Same-objective Stage 2 continuation can refine deterministic fidelity, but it is unlikely to produce the visibly new fine texture expected from a generative model.

## 7 Stage 2 XL Candidate Selection

The XL condition encoder uses:

```
config: configs/latent_pretrain_photo100k_v3_noise_xl.yaml
base_channels: 256
num_blocks: 16
degradation: photo_v3_noise_mix
finished step: 80000
```

Candidate comparison on the same validation images:

| Candidate        | Step  | Latent loss | Latent MSE | Decoded PSNR |
|------------------|-------|-------------|------------|--------------|
| best_eval_latent | 66000 | 0.27230     | 0.91770    | 21.3828      |
| step_0072000     | 72000 | 0.27940     | 0.97295    | 21.5241      |
| latest           | 80000 | 0.27593     | 0.89609    | 21.5062      |

The 72k checkpoint was selected for Stage 4 XL because it had the best decoded condition-only PSNR on the fixed validation set.

## 8 Stage 4 XL Edge-Loss Result

The first XL diffusion run used the selected Stage 2 XL condition encoder and partial initialization from the smaller Stage 4 v2 checkpoint.

```
config: configs/diffusion_photo100k_xl_stage4_condition_v3_edge_b16.yaml
run: diffusion_photo100k_xl_stage4_condition_v3_edge_b16
U-Net path: 469.6M parameters
full inference path: 509.658M parameters
train batch size: 16
GPUs: 2x A100-SXM4-80GB through PyTorch DDP
finished step: 5000
selected checkpoint: step 4250, best eval/decoded_mse
```

Training-time best proxy:

```
step 4250
eval/decoded_psnr: 21.9872
eval/decoded_mse: 0.025313
eval/noise_mse: 27.66008
eval/x0_mse: 0.86226
```

Sampled validation evaluation:

```
checkpoint: best_eval_condition_decoded.pt
checkpoint step: 4250
split: val
limit: 100
init: condition
start_timestep: 50
steps: 32
mean_bicubic_psnr: 22.3599
mean_sr_psnr: 23.0793
mean_psnr_delta: +0.7195
```

The latest XL Stage 4 checkpoint is therefore better than bicubic on the current v3 validation setup and better aligned with the desired denoise/color cleanup behavior than the Stage 2 condition-only path. It is still not a final restoration model: outputs remain softer than GT on fine textures.

## 9 Stage 4 XL Role-Split and Gated-Residual Probes

Follow-up probes tested whether Stage 4 could act as a bounded detail refiner instead of freely overwriting the deterministic Stage 2 condition latent.

The role-split mild probe added a lowpass anchor and separated low/high-frequency decoded losses:

```
config: configs/diffusion_photo100k_xl_stage4_condition_v3_rolesplit_mild_b8_probe.yaml
run: diffusion_photo100k_xl_stage4_condition_v3_rolesplit_mild_b8_probe
W&B: https://wandb.ai/jwheo/LuSIR/runs/lrb6nco9
finished step: 8000 micro-steps, 2000 optimizer updates
best one-step checkpoint: step 7500, decoded PSNR 23.2515
```

Sampled mild val100 showed that role-split losses reduced some over-editing at very low start timesteps, but the diffusion path still damaged the Stage 2 condition output on average:

| Model                     | Start timestep | Mean SR PSNR | vs condition-only | Wins vs condition |
|---------------------------|----------------|--------------|-------------------|-------------------|
| Stage 2 XL condition-only | n/a            | 25.0449      | n/a               | n/a               |
| Stage 4 role-split        | 25             | 24.5747      | -0.4702           | 3/100             |
| Stage 4 role-split        | 10             | 24.9185      | -0.1264           | 3/100             |
| Stage 4 role-split        | 5              | 24.9935      | -0.0514           | 6/100             |
| Stage 4 role-split        | 1              | 25.0335      | -0.0114           | 10/100            |

The next probe changed the diffusion prediction parameterization to `gated_residual_x0`. Instead of predicting the full denoised latent, the U-Net predicts a bounded residual and gate on top of the condition latent:

```
x0_hat = condition + residual_scale * tanh(residual_logits) * sigmoid(gate_logits + gate_bias)
residual_scale: 1.25
gate_bias: 0.0
out_channels: 32
```

The probe was initialized from the role-split best checkpoint with partial initialization and stopped at step 2000 after the sampled validation result reached condition-only parity:

```
config: configs/diffusion_photo100k_xl_stage4_condition_v3_gated_residual_mild_b8_probe.yaml
run: diffusion_photo100k_xl_stage4_condition_v3_gated_residual_mild_b8_probe
W&B: https://wandb.ai/jwheo/LuSIR/runs/edfko8e8
best one-step checkpoint: step 1000
final checkpoint: step 2000
```

Sampled mild val100:

| Model                     | Checkpoint | Start timestep | Mean SR PSNR | vs condition-only | Wins vs condition |
|---------------------------|------------|----------------|--------------|-------------------|-------------------|
| Stage 2 XL condition-only | n/a        | n/a            | 25.0449      | n/a               | n/a               |
| Stage 4 gated residual    | 1000       | 25             | 25.0415      | -0.0035           | 25/100            |
| Stage 4 gated residual    | 1000       | 10             | 25.0415      | -0.0034           | 25/100            |
| Stage 4 gated residual    | 1000       | 5              | 25.0416      | -0.0034           | 25/100            |
| Stage 4 gated residual    | 1000       | 1              | 25.0418      | -0.0032           | 25/100            |
| Stage 4 gated residual    | 2000       | 25             | 25.0445      | -0.0004           | 34/100            |
| Stage 4 gated residual    | 2000       | 10             | 25.0444      | -0.0006           | 32/100            |
| Stage 4 gated residual    | 2000       | 5              | 25.0444      | -0.0005           | 31/100            |
| Stage 4 gated residual    | 2000       | 1              | 25.0443      | -0.0007           | 32/100            |

This is a partial success. The gated residual parameterization almost eliminated the condition damage seen in previous Stage 4 probes and increased the number of samples beating condition-only from 10/100 at role-split t1 to 34/100 at gated-residual step2000 t25. However, the mean result still does not beat the Stage 2 condition-only baseline. The current interpretation is that Stage 4 has learned a safe near-identity residual path, not a reliable missing-detail generator.

## 10 Stage 2 Residual Diagnostic and Deterministic Refiner

A direct residual/oracle diagnostic was run to separate the Stage 2 condition-only error into lowpass and highpass components on mild val100:

| Metric                         | Value    |
|--------------------------------|----------|
| Bicubic PSNR                   | 24.4778  |
| Condition decoded PSNR         | 25.0543  |
| Oracle full residual PSNR      | 41.8207  |
| Oracle full vs condition       | +16.7664 |
| Oracle highpass PSNR           | 35.0872  |
| Oracle highpass vs condition   | +10.0329 |
| Oracle lowpass PSNR            | 25.0814  |
| Oracle lowpass vs condition    | +0.0270  |
| Residual highpass energy ratio | 0.8988   |
| Residual lowpass energy ratio  | 0.0758   |

This indicates that the Stage 2 condition encoder already preserves most structure and low-frequency color, while the remaining recoverable error is dominated by high-frequency detail.

The follow-up deterministic bounded residual refiner freezes the Stage 1 VAE and Stage 2 condition encoder, then predicts only:

`condition + residual_scale * tanh(residual_logits) * sigmoid(gate_logits + gate_bias)`

The sparse-gate probe reached its best validation result at step 500:

| Model                        | Mean PSNR | vs condition | Wins vs condition | Gate mean |
|------------------------------|-----------|--------------|-------------------|-----------|
| Stage 2 condition-only       | 25.0449   | n/a          | n/a               | n/a       |
| Sparse-gate residual refiner | 25.1178   | +0.0729      | 86/100            | 0.2147    |
| Open-gate residual refiner   | 25.0972   | +0.0523      | 73/100            | 0.8680    |

The sparse-gate result is a small but real gain and is qualitatively close to the condition output, without the destructive edits seen in previous diffusion Stage 4 probes. The open-gate ablation was worse despite a much larger mean gate value, so the next step should not simply force larger residual edits.

The sparse-gate refiner was then connected to standalone eval and inference scripts and evaluated without retraining across the active degradation presets:

| Degradation        | Bicubic PSNR | Condition PSNR | Refined PSNR | vs condition | Wins   |
|--------------------|--------------|----------------|--------------|--------------|--------|
| mild               | 24.4778      | 25.0449        | 25.1178      | +0.0729      | 86/100 |
| photo_v2           | 22.4103      | 22.9271        | 22.9767      | +0.0496      | 77/100 |
| photo_v3_noise_mix | 22.3599      | 22.9014        | 22.9600      | +0.0586      | 86/100 |

Qualitatively, the refined outputs remain very close to the Stage 2 condition outputs. This is useful because the refiner does not introduce the destructive edits seen in earlier diffusion probes, but the visible detail gain is still small. The result is best interpreted as a safe residual teacher or warm start, not as a final detail generator.

## 11 Teacher-Supervised Stage 4 Residual Probe

The sparse-gate deterministic refiner was used as a frozen teacher for the gated-residual Stage 4 U-Net. Direct losses supervised the predicted residual, highpass residual, and gate on `photo_v3_noise_mix`.

config: configs/diffusion\_photo100k\_xl\_stage4\_condition\_v3\_teacher\_residual\_photo\_v3\_b8\_probe.yaml  
train batch size: 8  
gradient accumulation: 4  
finished step: 8000 micro-steps, 2000 optimizer updates

Sampled val100, condition initialization, 32 sampling steps:

| Checkpoint                      | Start timestep | Mean SR PSNR | vs bicubic | vs condition | Wins vs condition |
|---------------------------------|----------------|--------------|------------|--------------|-------------------|
| Teacher Stage 4 step 2000       | 25             | 22.9640      | +0.6041    | +0.0626      | 68/100            |
| Teacher Stage 4 step 2000       | 50             | 22.9639      | +0.6040    | +0.0625      | n/a               |
| Teacher Stage 4 step 4000       | 25             | 22.9571      | +0.5972    | +0.0557      | 65/100            |
| Teacher Stage 4 step 8000       | 25             | 22.9490      | +0.5891    | +0.0476      | 59/100            |
| Existing edge Stage 4 step 4250 | 25             | 22.9563      | +0.5964    | +0.0549      | 42/100            |
| Existing edge Stage 4 step 4250 | 50             | 23.0799      | +0.7200    | +0.1784      | 45/100            |

Teacher supervision therefore produced a small, stable cleanup gain over the Stage 2 condition output and slightly exceeded the edge model at t25. However, the user-facing objective was not achieved. Fur, leaves, branches, and building detail remained strongly smoothed. A simple absolute-Laplacian diagnostic measured teacher step-2000 output at 21.8% of GT detail energy, below the existing edge t25 output at 32.7%. This is not a complete perceptual metric, but it agrees with visual inspection.

The best sampled checkpoint was step 2000; continuing to step 8000 reduced both mean PSNR and condition win count. The active interpretation is that the current `photo_v3_noise_mix` curriculum overemphasizes severe denoise/cleanup cases, so direct residual supervision teaches a safer cleanup operator rather than a missing-detail generator.

## 12 Detail-Preserving Curriculum Adaptation

A fixed-sample degradation audit confirmed the curriculum mismatch. On val100, `photo_v3_noise_mix` reduced the bicubic baseline to 22.3599 PSNR and produced mean LR chroma RMS error 0.02040 relative to clean downsampling. `photo_detail_mix` increased the bicubic baseline to 24.7357 and reduced the chroma error to 0.00507.

The existing Stage 2 XL condition encoder was evaluated before retraining. It already improved `photo_detail_mix` from 24.7357 bicubic PSNR to 25.3103, a +0.5745 gain. Stage 2 was therefore frozen, and the teacher-supervised gated-residual Stage 4 was adapted from its selected step-2000 checkpoint:

config: configs/diffusion\_photo100k\_xl\_stage4\_condition\_v3\_teacher\_residual\_photo\_detail\_b8\_long.yaml  
run: diffusion\_photo100k\_xl\_stage4\_condition\_v3\_teacher\_residual\_photo\_detail\_b8\_long  
W&B: <https://wandb.ai/jwheo/LuSIR/runs/so0lbyte>  
train batch size: 8  
gradient accumulation: 4  
learning rate: 1e-6  
finished step: 12000 micro-steps, 3000 optimizer updates

Sampled `photo_detail_mix` val100, condition initialization, start timestep 25, 32 sampling steps:

| Model/checkpoint                | Mean SR PSNR | vs bicubic | vs condition | Wins vs condition |
|---------------------------------|--------------|------------|--------------|-------------------|
| Stage 2 condition-only          | 25.3103      | +0.5745    | n/a          | n/a               |
| Teacher Stage 4 initialization  | 25.3187      | +0.5829    | +0.0084      | 46/100            |
| Photo-detail Stage 4 step 8000  | 25.3406      | +0.6049    | +0.0303      | 71/100            |
| Photo-detail Stage 4 step 12000 | 25.3337      | +0.5980    | +0.0235      | 67/100            |
| Existing edge Stage 4 step 4250 | 25.1176      | +0.3818    | -0.1927      | 13/100            |

Step 8000 is the selected checkpoint. This is the first sampled gated-residual Stage 4 result that beats the Stage 2 condition-only baseline on both mean PSNR and a clear majority of samples. Qualitatively, it preserves the



condition structure and sharpness and avoids the broad destructive edits produced by the edge model on this distribution.

The result remains conservative. Mean absolute-Laplacian energy is 29.7% of GT for step 8000, compared with 29.6% for its teacher initialization and 41.2% for the more aggressive edge model. The gain therefore comes primarily from more accurate bounded corrections, not from strong new texture synthesis. Rare strong-tail samples can still exhibit bright artifacts.

## 13 Decoded-Detail Residual Refiner v2

The deterministic bounded residual refiner uses 192 hidden channels, 12 residual blocks, and decoded-image/highpass supervision through the frozen VAE decoder. An initial 12000-step run was continued to 40000 micro-steps with a lower  $2.5\text{e-}5$  learning rate. Step 39000 achieved the best global decoded PSNR and was selected.

| Degradation                     | Condition mean PSNR | Refined mean PSNR | Gain    | Wins   |
|---------------------------------|---------------------|-------------------|---------|--------|
| <code>photo_detail_mix</code>   | 25.3103             | 25.6410           | +0.3307 | 94/100 |
| <code>mild</code>               | 25.0449             | 25.3161           | +0.2712 | 91/100 |
| <code>photo_v2</code>           | 22.9271             | 23.0419           | +0.1148 | 81/100 |
| <code>photo_v3_noise_mix</code> | 22.9014             | 23.0787           | +0.1773 | 81/100 |

On the training curriculum, step 39000 reached global decoded PSNR 24.0305, +0.2927 dB over condition-only, mean PSNR gain +0.3307 dB, SSIM gain +0.01076, and wins on 94/100 images. Step 40000 had a slightly higher SSIM gain (+0.01161) but lower PSNR and fewer wins, so step 39000 is the more balanced public default.

The continuation improved mean PSNR on every tested degradation. However, the strong `photo_v2` and `photo_v3_noise_mix` win counts fell versus step 11000, indicating that the larger correction has a less conservative failure tail. A post-training residual-strength guardrail provides an explicit trade-off: strength 1.0 gives the best average quality, 0.75 preserves most of the gain with fewer regressions, and 0.5 raises strong-preset wins from 81/100 to 86/100 while keeping positive mean gains. These modes are exposed in the inference CLI and Colab. Future work should prioritize user-facing evaluation and a learned degradation-aware gate rather than continuing indefinitely.

## 14 Systems Notes

Diffusion training now supports PyTorch DDP when launched with `torchrun`. Without `torchrun`, the same script falls back to the existing single-GPU path. On the tested 2x A100 SXM environment, a 1 GiB NCCL all-reduce smoke test reported about 199.5 GB/s, so multi-GPU communication was not the bottleneck.

The completed Stage 4 XL edge run trained at about 0.78 step/s in ordinary train sections. A 5000-step run is roughly 1.8 hours of pure training, plus eval/checkpoint overhead.

On the tested 2x L40S environment, the gated-residual probe ran with high GPU utilization, about 0.79 micro-step/s, and roughly 44.7 / 46.1 GiB allocated per GPU. No GPU bottleneck or NCCL communication issue was observed.

The decoded-detail residual refiner v2 ran on one L40S at about 0.89 micro-step/s with 42.0 / 46.1 GiB allocated and sustained 99–100% GPU utilization.

## 15 Visual Review and External Positioning

A fixed-sample visual review now compares LR, bicubic, Stage 2 condition, residual strengths 0.50, 0.75, and 1.00, and GT side by side. On mild inputs the selected refiner makes small, generally structure-preserving improvements. On stronger `photo_v2` and `photo_v3_noise_mix` inputs, the main failure occurs earlier: the condition encoder removes noise together with real detail and sometimes leaves small cyan/white grid-like artifacts. The residual refiner changes are too small to recover the missing fur, leaf veins, text, and distant structural detail.

Qualitatively, the current model is not competitive with leading generative restoration systems in perceived sharpness or plausible fine texture. Its useful distinction is a deterministic, vision-only path trained without a pretrained

text-to-image model, with lower hallucination risk and adjustable correction strength. This is an architectural and visual assessment, not a direct SOTA benchmark. A defensible comparison requires same-input blind A/B testing plus LPIPS/DISTS/MANIQA/MUSIQ and user-preference evaluation.

### 15.1 Representative Visual Examples

The following two examples use the same lime image so that the role of the base reconstruction and the later detail branch can be inspected separately. They are representative diagnostic examples, not cherry-picked evidence of a formal benchmark win.



Figure 1. Multiscale Stage 2 restores the degraded input while retaining a visible fine-texture gap to ground truth.



Figure 2. Selected detail branch v1d adds a controlled, artifact-light correction over the base; this illustrates both its improvement and its remaining conservative behavior.

## 16 Stage 2 Multiscale Redesign

A decoded pixel/edge/highpass fine-tuning probe confirmed that the flat Stage 2 condition predictor is the current bottleneck. By step 4000, decoded PSNR had risen from 23.7387 to 24.2792, but the Laplacian detail ratio fell from 0.2817 to 0.2773 and fixed samples remained visibly smooth. This is consistent with the distortion-perception limitation of optimizing reconstruction losses without sufficient spatial context or high-quality training exposure.

The replacement keeps all parameters and names of the selected 19M-parameter step 72000 predictor, then adds a zero-output-initialized 128-to-64-to-32 multiscale context branch. Partial initialization therefore exactly preserves the previous output before training. The training manifest also changes from 100,000 COCO plus 3,450 DIV2K/Flickr2K rows to a roughly balanced exposure of 100,000 COCO and 103,500 repeated DIV2K/Flickr2K rows. Random crops and degradations remain stochastic, so repeated high-quality rows generate different training pairs.

The design is informed by broad-context and multiscale restoration findings in [SwinIR](#), [HAT](#), and [NAFNet](#), together with the data/degradation emphasis of [Real-ESRGAN](#). The resulting 55.50M-parameter model passed a full batch-8 forward/backward and val100 smoke test on one L40S at approximately 34.8/46.1GB VRAM and 100% GPU utilization. The 50,000-micro-step long run is tracked at <https://wandb.ai/jwheo/LuSIR/runs/6zt2do4v>.

The run completed normally. Step 46000 was selected over the final checkpoint because it provides the best cross-preset distortion/detail compromise. It improves the previous Stage 2 by +1.0348 dB on `photo_detail_mix` and +0.9228 dB on `mild`, with 99/100 and 97/100 per-image wins and slightly higher detail ratios. On the stronger `photo_v2` and `photo_v3_noise_mix` presets it improves PSNR by +0.9441 dB and +0.9650 dB, but the detail ratio drops from roughly 0.29-0.30 to 0.22-0.23. Visual comparison confirms that the model improves color, large boundaries, and denoising without convincingly recovering missing fur, text, fabric, or distant texture. The experiment is a successful base-reconstruction redesign but not a solution to perceptual fine-detail recovery.

An optional continuation completed from step 46000 using frozen ImageNet VGG16 features at shallow and intermediate layers. This explicitly introduces pretrained vision feature supervision, while still avoiding pretrained text-to-image or generative models. A CUDA smoke test passed at batch 4 with approximately 20.6/46.1GB VRAM and 2.62 micro-steps/s. The run completed 12,000 micro-steps. Step 8000 had the best shortlist score, 26.0092, and improved decoded PSNR over initialization by +0.0101 to +0.0256 dB across the four tested presets. Step 11000 was strongest on cleaner presets but regressed `photo_v3_noise_mix` by -0.0063 dB. Fixed contact sheets showed almost no visible difference from initialization, and missing fine texture remained smoothed. The run is preserved as a partial metric/latent success but is not promoted into the public path. The completed run is tracked at <https://wandb.ai/jwheo/LuSIR/runs/nrqhw05u>.

## 17 Dual-Context Unique-Data Stage 2 Scale-up

The VGG continuation's +0.0101 to +0.0256 dB cross-preset improvements were not visually distinguishable and did not recover missing texture. Capacity and data uniqueness are therefore changed together instead of extending the same objective again. The previous HQ-balanced manifest had 203,600 rows but only 103,550 unique images because DIV2K and Flickr2K were repeatedly exposed.

The new manifest adds 30,000 unique LSDIR images for 133,450 unique training images plus the unchanged 100-image validation set. The predictor retains the selected 55.50M-parameter multiscale model and adds a second zero-output- initialized multiscale context branch. Partial initialization exactly preserves the selected step 46000 output before training while increasing capacity to 119.24M parameters.

The one-L40S smoke test reproduced the initial 24.48 dB decoded PSNR and 0.291 detail ratio. Batch 8 with gradient accumulation 4 used approximately 37.8/46.1GB VRAM, reached 99% GPU utilization, and sustained approximately 0.75 micro-step/s. The completed long run targeted 100,000 micro-steps, or 25,000 optimizer updates. Evaluation ran every 1,000 micro-steps, while ordinary milestone checkpoints were limited to every 5,000 micro-steps to stay within the disk budget. The run is tracked at <https://wandb.ai/jwheo/LuSIR/runs/4akqckxu>.

The run completed all 100,000 micro-steps. Step 98,000 is the automatic `eval/decoded_psnr` best checkpoint on `photo_detail_mix`; step 100,000 is slightly better on stronger degradations. Re-evaluated with the same comparison tool against selected step 46,000, the gains are +0.1362 dB on `photo_detail_mix`, +0.1086 dB on `mild`, +0.0540 dB on `photo_v2`, and -0.0356 dB on `photo_v3_noise_mix` for the best checkpoint. The final checkpoint changes those to +0.1256, +0.1025, +0.0668, and +0.0132 dB. This is a modest reconstruction improvement, not a solved perceptual-detail problem.

## 18 High-Frequency Detail Branch v1

The latest deterministic detail candidate is an image-space detail branch, not another Stage 4 continuation. The path is:

```
LR -> frozen Stage 2 dual-context condition -> frozen Stage 1 decoder -> base SR
base SR + bicubic LR upsample -> gated high-frequency detail branch -> detail SR
```

The branch predicts a bounded RGB residual and a per-pixel gate. The residual is projected through a local highpass operation before being added to the base SR, which limits the model's ability to change global color or

low-frequency structure. The output convolution is zero-initialized, so step 0 exactly reproduces the frozen Stage 2 + Stage 1 base output.

Implemented files:

```
configs/detail_branch_v1_photo130k_lsdire.yaml
configs/detail_branch_v1b_aug_photo130k_lsdire.yaml
configs/detail_branch_v1c_condition_open_photo130k_lsdire.yaml
configs/detail_branch_v1d_deep3m_photo130k_lsdire_3ep.yaml
configs/hf/detail_branch_v1d_deep3m_photo130k_lsdire_3ep.yaml
tools/train/train_detail_branch.py
tools/eval/run_fixed_review_detail_branch.py
tests/test_detail_branch.py
```

A four-micro-step smoke test completed load/eval/backprop/update/checkpoint successfully. Step 0 matched the base output exactly at 24.6188 dB val100 PSNR; after one optimizer update, the branch produced a tiny +0.00005 dB aggregate PSNR delta and 69/100 wins versus base. This is only an integration check, not a quality claim.

The first v1 run was stopped at 7800 micro-steps, or 0.234 epoch over the 133450-image train manifest, because early samples showed that the residual was safe but visually too small. The v1b run kept the same model and loss, but added horizontal flips, texture-biased crop retry, and weak HR color jitter while excluding rotation, vertical flip, affine/perspective transforms, erasing, and mixup.

The v1b run completed 40000 micro-steps, equal to 10000 optimizer updates with `grad_accum_steps: 4`. It selected step 39500 by `eval/detail_score`:

|                  |            |
|------------------|------------|
| base PSNR:       | 24.6188    |
| detail PSNR:     | 24.6649    |
| PSNR delta:      | +0.0461 dB |
| base SSIM:       | 0.80013    |
| detail SSIM:     | 0.80281    |
| SSIM delta:      | +0.00268   |
| mean PSNR delta: | +0.0575    |
| wins vs base:    | 98/100     |
| detail wins:     | 100/100    |

Different metrics peak at nearby checkpoints: PSNR delta is highest at step 38500 (+0.0489 dB), SSIM delta is highest at step 37000 (+0.00336), and the final step 40000 reaches +0.0444 dB PSNR and +0.00277 SSIM with 98/100 wins. The selected step 39500 remains preserved as the earlier public detail artifact because it has the best combined detail score from the completed v1b run. Qualitatively, it is artifact-light and slightly sharper on texture-heavy crops, but still conservative and below GT on fine surface detail. It remains a historical comparison artifact; the later selected v1d checkpoint is exposed through the Colab WebUI detail runner.

V1c initialized from the selected v1b checkpoint, exposed the frozen Stage 2 condition latent directly to the image-space branch, increased the bounded residual scale, and opened the gate slightly. The selected step 6000 improves the fixed `photo_detail_mix` val100 base by +0.0554 dB aggregate PSNR and +0.00332 SSIM with 99/100 wins. This was better than v1b but plateaued quickly enough to motivate a controlled capacity test.

V1d keeps the v1c width and objective but increases residual depth from 8 to 18 blocks, raising branch capacity from 1.35M to 3.02M parameters. The original blocks are copied from v1c step 6000 and the ten appended blocks are identity-initialized, so the step-0 output exactly reproduces v1c. The run completed 100,086 micro-steps, exactly three passes over the 133,450-image manifest at batch size 4, and selected step 99,500 by `eval/detail_score`.

On ordinary fixed `photo_detail_mix` val100, selected step 99,500 improves the frozen base by +0.1646 dB aggregate PSNR, +0.1888 dB mean PSNR, and +0.00647 SSIM with 99/100 PSNR wins and 100/100 detail wins. On the strict-bicubic five-crop diagnostic it reaches 31.9513 dB, improving the frozen base by +0.2102 dB, v1c by +0.1358 dB, and winning 5/5 images. This is also +0.1266 dB above the early v1d step-9500 snapshot.

Final step 100,086 reaches a nearly identical strict-bicubic 31.9516 dB, but selected step 99,500 is stronger on ordinary-val aggregate PSNR, SSIM, highpass improvement, and the combined detail score. Visual inspection shows no white-dot, grid, or excessive-sharpening artifacts. The larger branch and long run therefore produced

meaningful stable progress, but still did not close the visible fine-texture gap to ground truth. Further same-objective continuation or simple capacity scaling is not planned.

## 19 Public Artifacts

The latest LuSIR public artifacts are stored in [jwheo/LuSIR](#) on Hugging Face:

```
checkpoints/stage4_photo100k_xl_edge_b16_best_eval_condition_decoded.pt
checkpoints/residual_refiner_stage2_xl_mild_best_eval_refined.pt
checkpoints/stage4_photo100k_xl_teacher_residual_photo_v3_step_0002000.pt
checkpoints/stage4_photo100k_xl_teacher_residual_photo_detail_best8000.pt
checkpoints/residual_refiner_stage2_xl_photo_detail_v2_best39000.pt
checkpoints/stage2_photo100k_multiscale_hqmix_step_0046000.pt
checkpoints/stage2_photo100k_multiscale_hqmix_perceptual_step_0008000.pt
checkpoints/stage2_photo130k_lsdir_dual_multiscale_best98000.pt
checkpoints/detail_branch_v1b_aug_photo130k_lsdir_best39500.pt
checkpoints/detail_branch_v1d_deep3m_photo130k_lsdir_best99500.pt
metrics/stage4_photo100k_xl_edge_b16_val100_t50_32step_summary.json
metrics/diagnose_stage2_xl_residuals_mild_val100_summary.json
metrics/residual_refiner_stage2_xl_mild_probe_early_stop_summary.json
metrics/eval_residual_refiner_stage2_xl_photo_v3_noise_mix_val100_summary.json
metrics/stage4_photo100k_xl_teacher_residual_photo_v3_step2000_val100_t25_32step_summary.json
metrics/stage4_photo100k_xl_teacher_residual_photo_v3_step2000_val100_t50_32step_summary.json
metrics/stage4_photo100k_xl_teacher_residual_photo_detail_best8000_val100_t25_summary.json
metrics/residual_refiner_stage2_xl_photo_detail_v2_long_summary.json
metrics/residual_refiner_v2_best39000_strength_sweep_summary.json
metrics/eval_residual_refiner_v2_best39000_stage2_xl_mild_val100_summary.json
metrics/eval_residual_refiner_v2_best39000_stage2_xl_photo_v2_val100_summary.json
metrics/eval_residual_refiner_v2_best39000_stage2_xl_photo_v3_noise_mix_val100_summary.json
metrics/stage2_multiscale_hqmix_step46000_cross_preset_summary.json
metrics/stage2_multiscale_perceptual_photo_detail_mix_candidates.json
metrics/stage2_multiscale_perceptual_mild_candidates.json
metrics/stage2_multiscale_perceptual_photo_v2_candidates.json
metrics/stage2_multiscale_perceptual_photo_v3_noise_mix_candidates.json
metrics/stage2_photo130k_lsdir_dual_multiscale_final_summary.json
metrics/detail_branch_v1b_aug_photo130k_lsdir_summary.json
metrics/detail_branch_v1d_deep3m_photo130k_lsdir_3ep_summary.json
metrics/benchmark_bicubic5_detail_v1d_best99500_summary.json
samples/stage4_photo100k_xl_edge_b16_val100_t50_32step_grid_lr_bicubic_sr_gt.png
samples/diagnose_stage2_xl_residuals_mild_val100_grid.png
samples/residual_refiner_stage2_xl_mild_probe_step500_grid.png
samples/eval_residual_refiner_stage2_xl_photo_v3_noise_mix_val100_grid.png
samples/compare_residual_refiner_vs_stage4_edge_0801_photo_v3.png
samples/stage4_photo100k_xl_teacher_residual_photo_v3_step2000_val100_t25_grid.png
samples/stage4_photo100k_xl_teacher_residual_photo_detail_best8000_val100_t25_grid.png
samples/residual_refiner_stage2_xl_photo_detail_v2_best39000_grid.png
samples/eval_residual_refiner_v2_best39000_stage2_xl_mild_val100_grid.png
samples/eval_residual_refiner_v2_best39000_stage2_xl_photo_v2_val100_grid.png
samples/eval_residual_refiner_v2_best39000_stage2_xl_photo_v3_noise_mix_val100_grid.png
samples/stage2_multiscale_hqmix_checkpoint_comparison.png
samples/stage2_multiscale_perceptual_photo_detail_mix_candidates.png
samples/stage2_multiscale_perceptual_photo_v3_noise_mix_candidates.png
samples/stage2_dual_lsdir_photo_detail_mix_best98k_final100k_contact_sheet.png
samples/stage2_dual_lsdir_mild_best98k_final100k_contact_sheet.png
samples/stage2_dual_lsdir_photo_v2_best98k_final100k_contact_sheet.png
samples/stage2_dual_lsdir_photo_v3_noise_mix_best98k_final100k_contact_sheet.png
```

```

samples/detail_branch_v1b_aug_photo130k_lsdire_best39500_grid.png
samples/detail_branch_v1d_deep3m_photo130k_lsdire_best99500_grid.png
samples/benchmark_bicubic5_detail_v1d_best99500_grid.png
configs/diffusion_photo100k_xl_stage4_condition_v3_edge_b16.yaml
configs/residual_refiner_stage2_xl_mild_probe.yaml
configs/diffusion_photo100k_xl_stage4_condition_v3_teacher_residual_photo_v3_b8_probe.yaml
configs/diffusion_photo100k_xl_stage4_condition_v3_teacher_residual_photo_detail_b8_long.yaml
configs/residual_refiner_stage2_xl_photo_detail_v2_continue_40k.yaml
configs/hf/residual_refiner_stage2_xl_photo_detail_v2.yaml
configs/latent_pretrain_photo100k_multiscale_hqmix_perceptual_continue.yaml
configs/latent_pretrain_photo130k_lsdire_dual_multiscale_long.yaml
configs/detail_branch_v1_photo130k_lsdire.yaml
configs/detail_branch_v1b_aug_photo130k_lsdire.yaml
configs/hf/detail_branch_v1b_aug_photo130k_lsdire.yaml
configs/detail_branch_v1d_deep3m_photo130k_lsdire_3ep.yaml
configs/hf/detail_branch_v1d_deep3m_photo130k_lsdire_3ep.yaml

```

## 20 Next Work

The selected residual refiner remains the public Colab default, while detail branch v1d is available as a selectable single-image/tiled Colab research option. The completed perceptual Stage 2 continuation is not promoted. Candidate next steps are:

- preserve the deterministic base as the fidelity path and train a separate gated, bounded high-frequency residual diffusion model for stochastic texture synthesis;
- select the generative path with LPIPS, DISTS, fixed visual review, high-frequency metrics, lowpass drift, and seed diversity instead of PSNR alone;
- continue Stage 2/base architecture research as a separate clean-fidelity track for the measured 0.9235 dB DIV2K gap, rather than mixing that goal into the generative detail objective;
- because v1d remains visually conservative, consider detail-only perceptual or adversarial supervision instead of increasing branch capacity again;
- keep a degradation-aware gate or strong-input guardrail as the primary response to the remaining strong-preset failure tail.

The first residual diffusion probe is specified in `docs/HIGH_FREQUENCY_RESIDUAL_DIFFUSION_KO.md`.